Which drug–hearing-loss signals in FAERS survive adjustment and independent replication? A label-referenced, externally replicated disproportionality study

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Key points

1 Introduction

Drug-induced hearing loss is usually irreversible, and the classical ototoxic drug classes—aminoglycosides, platinum agents, loop diuretics, antimalarials, macrolides and salicylates—were identified clinically long before spontaneous-reporting databases existed [1]. For newer drugs, post-marketing spontaneous reports are often the first source of evidence, and several recent studies have screened the FDA Adverse Event Reporting System (FAERS) for hearing-loss signals across all drugs [2–4] and for tinnitus [5].

Those screens share a design: crude disproportionality across all reports, with novelty defined as absence from the label. Crude disproportionality in this setting is vulnerable to at least three biases that are specific to hearing outcomes. First, confounding by indication and protopathic bias: corticosteroids, vasodilators and antivirals are prescribed *for* sudden hearing loss, tinnitus and vestibular disorders, so they co-occur with the event in reports for reasons unrelated to toxicity. Second, the "innocent bystander" effect: a drug that is routinely co-administered with a strongly ototoxic one—an antiemetic given with cisplatin, an antibiotic given with an aminoglycoside—inherits part of the association [6]. Third, stimulated reporting after safety communications or litigation [7]. In addition, crude screens rarely test whether their signal list agrees with an independent data source, and editors have begun to reject FAERS analyses that lack such validation [8].

We therefore asked a narrower question than "which drugs are ototoxic?": *which hearing-loss and tinnitus signals remain after adjustment for these biases, and do the signals that remain replicate in an independent national database and over time?* We evaluated crude and adjusted detection against a reference set derived from current US labels, replicated every signal in Canada Vigilance, and compared our results with the signal lists of previous FAERS screens. The analysis protocol, including all thresholds, was committed to a public repository before the first result was computed.

2 Methods

Reporting follows the READUS-PV guideline for disproportionality analyses [9]; the checklist is provided as Supplementary Table S1.

2.1 Data sources

We used all 90 quarterly ASCII extracts of the legacy Adverse Event Reporting System (2004Q1–2012Q3) and FAERS (2012Q4–2026Q2), the Canada Vigilance Adverse Reaction Online Database extract covering 1965 to 30 June 2026, and the openFDA bulk export of structured product labels (23 October 2026) [10]. All data are public and de-identified; no ethics approval was required.

2.2 Pre-processing and deduplication

In FAERS, we kept the most recent version of each case (by FDA receipt date, preferring FAERS over legacy records, then case version), removed all case identifiers that FDA lists as deleted, and collapsed content duplicates: reports with identical sex, age, event date, country, set of suspect ingredients and set of reactions (applied only when event date and age were present). Drug names were mapped to active ingredients: the `prod_ai` field was split into ingredients, salt and hydrate words were removed when the remaining base was itself an observed ingredient, and free-text names in records without `prod_ai` were mapped to the dominant ingredient seen for that name ($\geq$70% of $\geq$5 records). In Canada Vigilance, reports flagged as duplicates of a report present in the extract were removed, and product ingredients were normalised with the same rules.

2.3 Outcomes and exposure

The primary outcome, hearing loss, comprised the MedDRA preferred terms Deafness; Deafness unilateral, bilateral, neurosensory, transitory, permanent and mixed; Hypoacusis; Neurosensory hypoacusis; Sudden hearing loss; Ototoxicity; Hearing impaired; Hearing disability; and High frequency hearing loss. Conductive and congenital deafness were excluded. The secondary outcome was Tinnitus. An ingredient was considered exposed when reported as a suspect drug (FAERS primary suspect, secondary suspect or interacting; Canada suspect or interacting).

2.4 Disproportionality

For each ingredient we computed the reporting odds ratio (ROR) with a Woolf 95% confidence interval [11] and the information component (IC) with its lower credibility bound IC_{025} [12]. A *crude signal* required at least three cases, a lower ROR bound above 1 and $\mathrm{IC}_{025} > 0$.

The *adjusted* analysis was run in a restricted population and stratified. We excluded (i) reports in which any drug had an ear or hearing indication (deafness, hearing, tinnitus, ototoxicity, Ménière's disease, otitis, vertigo, vestibular, labyrinthine and cochlear terms), (ii) reports that mentioned, in any role, a drug from an a-priori list of 29 established ototoxic drugs [1] other than the drug under

study, and (iii) FAERS reports submitted by lawyers. We then computed the Mantel–Haenszel OR [13] over 60 strata (five age bands × sex [male, female, unknown] × four reporting periods) with the Robins–Breslow–Greenland variance [14]. An *adjusted signal* required at least three cases in the restricted population and a lower 95% bound above 1. The implementation was checked against `statsmodels` on simulated data (identical estimates to nine decimal places).

2.5 Reference set

From single-ingredient labels we classified each ingredient as label-positive when a hearing-loss term appeared in a safety section (boxed warning, warnings, precautions, adverse reactions, post-marketing experience, contraindications, overdosage, or the corresponding over-the-counter sections) of at least half of its prescription labels, and label-negative when no label mentioned hearing loss or tinnitus at all; other ingredients were left out. Phrases that contain a hearing word without describing an adverse effect (for example, the disease name "maternally inherited diabetes and deafness" and telephone lines for deaf users) were removed before matching. Discrimination was assessed by the area under the ROC curve (AUC) among ingredients with at least 1,000 suspect reports, comparing crude IC_{025} with the adjusted lower bound; the difference was bootstrapped (2,000 resamples). Because labels are partly informed by spontaneous reports, the reference set is used to compare methods with each other, not to estimate absolute accuracy.

2.6 Replication

For external replication, signals were derived in FAERS after removing reports originating in Canada and tested in Canada Vigilance. An ingredient was testable when the observed or expected number of Canadian cases was at least three, and replicated when the Canadian lower ROR bound exceeded 1 with at least three cases. The same rule was applied to label-negative ingredients that were not FAERS signals, giving the rate at which replication occurs without a true association. For temporal replication, signals derived from reports received up to 2014 were tested in reports received from 2015 onwards.

2.7 Robust signals and novelty

A *robust signal* was an adjusted FAERS signal that replicated in Canada. Robust signals without a hearing-loss term in any US label were checked against PubMed (drug name with hearing-loss or ototoxicity terms in title or abstract) and against the signal lists of the previous FAERS screens [2–5].

Declarations

Funding: none. **Conflicts of interest: Ethics:** not applicable (public, de-identified data). **Data and code availability:** all code, the pre-registered protocol and result tables are available at

Author contributions: OT conceived the study, ran the analysis and wrote the manuscript. **Use of AI tools:**

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